Converting Mass, Area and Volume

Teacher's Edition

The Period

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

Must not A ratio not on the FAST reference sheet (no fl oz → mL, no ft^2 → m^2 as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

Target I can convert mass, volume and area between systems, and square the length ratio for area.

MTR **MTR.5.1** — Notes II: yesterday's k^2 and today's 144 are the same structure, and the room is asked to say so · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — every mass and volume board says its estimate first

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Mass and volume convert like length: the sheet's ratio, the estimate first.
2. Square units, square the ratio: 1 ft^2 is 144 in^2, 1 m^2 is 10,000 cm^2.
3. Fluid ounces go through cups — read the sheet for the path.

Slide by Slide

1. **Converting Mass, Area and Volume.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes II: a square foot is 12 by 12 inches, 144 of them, not 12. *1 retrieves 5.06–07 — × the ratio whose 1 is inches (yesterday). 2 retrieves 5.02–03 — area × k^2; today's 144 is the same idea (earlier this unit). 3 retrieves 4.01 — a signed decimal product (last unit). 4 retrieves MA.6.GR.2.1 — 1 ft^2 is a 12 by 12 square; today's Notes II (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 30.48 cm 2. 63 cm^2 3. −10 4. 144 in^2
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the two lines.** 1 lb = 0.454 kg is a form of 1 like 1 in = 2.54 cm.
6. **Notes.** 1 ft = 12 in, but a square foot is a 12 by 12 square: 144 in^2.
7. **Notes.** The sheet has 1 c = 8 fl oz and 1 c = 0.24 L, and no line from fluid ounces to milliliters — so a fluid ounce reaches liters through cups: a 12-fl-oz can is 12 ÷ 8 = 1.5 c, and 1.5 × 0.24 = 0.36 L.
8. **Example 1 — Pounds to kilograms.** Ask which is heavier, a pound or a kilogram, before anyone multiplies.
9. **Example 1 — Worked.** 31.78 kg.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 40.86 kg — 40.9 to one place.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Square feet to square inches.** Ask how many times the ratio is used before anyone multiplies.
13. **Example 2 — Worked.** 720 in^2.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 45 ft^2.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $1\,\text{kg} = 2.204\,\text{lb}$. A bag of rice is 10 kg. How many pounds is that? → ***22.04 lb*** *4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added*

18–19. Board 2. $1\,\text{gal} = 3.785\,\text{L}$. A cooler holds 4 gal. How many liters is that? → ***15.14 L*** *1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added*

20–21. Board 3. $1\,\text{L} = 0.264\,\text{gal}$. A tank holds 10 L. How many gallons is that? → ***2.64 gal*** *37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10*

22–23. Board 4. $1\,\text{yd} = 3\,\text{ft}$. A patch of lawn is $2\,\text{yd}^2$ in area. How many square feet is that? → ***B — 18 ft^2*** *(A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) 3^2 alone: the 2 yd^2 forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. $1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. A bottle holds 16 fl oz. How many liters is that? → ***0.48 L*** *3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units]*

26–27. Board 6. $1\,\text{m} = 100\,\text{cm}$. How many square centimeters is $1\,\text{m}^2$? → ***C — 10,000 cm²*** *(A) 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $1\,\text{kg} = 2.204\,\text{lb}$. A 5-lb bag of flour and a 2.5-kg bag of sugar. Which is heavier, and by how much? → ***The sugar, by 0.51 lb*** *'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]; 'the flour, by 2.5 lb' — 5 − 2.5 in two units*

30–31. Board 8. $1\,\text{gal} = 3.785\,\text{L}$. A 20-gal fish tank, bought for $45, is being filled. How many liters does it hold? → ***75.7 L*** *The $45 is not needed. 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used*

32–33. Board 9. $1\,\text{ft} = 12\,\text{in}$. Priya says $2\,\text{ft}^3$ is $24\,\text{in}^3$, because $2 \times 12 = 24$. Name the error and fix it. → ***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in³, so 2 ft³ = 3,456 in³.*** *288 in³ — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–3 take one ratio each direction — mass, then volume both ways; 4 asks a square unit as a vote; 5 chains fluid ounces through cups. The pattern breaks at 6, where the familiar 100 is wrong. 7 converts and then compares, 8 is the extra number, 9 the cube in writing. The independent set walks the same path: mass both ways, volume, a square unit, fluid ounces through cups, square meters.

Misconceptions to Watch

Board 1. 4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]

Board 2. 1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

Board 3. 37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]

Board 4. 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]

Board 5. 3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]

Board 6. 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]

Board 7. 'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]

Board 8. 5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]

Board 9. 288 in^3 — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the

document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.